

DocIngestQA

Interview Defense Document v4.0

Pre-Indexing RAG Document Ingestion Auditing

Metric	Value
Checks	11 deterministic (8 in v0.1, 3 in v0.2)
Checks that can FAIL	3 (input structure, page coverage, encoding health)
Checks that WARN	8
Current version	v0.2.0
PyPI published	<code>pip install docingestqa</code>
Test suite	27/27 passing
Output formats	JSON Markdown HTML
CLI available	<code>python -m docingestqa audit</code>
Exit codes	PASS=0 WARN=1 FAIL=2 INSUFFICIENT_INPUT=3

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Sourced exclusively from repo files | All claims verified

Label Key

Label	Meaning
[IMPLEMENTED]	Code exists, runs, produces verified artifact or DB row
[SIMULATED]	Deliberately synthetic scenario; simulation logic is real code
[PARTIALLY VERIFIED]	Code exists but verification relies on internal consistency only
[FUTURE]	Not built -- explicitly absent from the repo

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Section 1 -- The Problem: Why RAG Pipelines Fail Silently

1.1 The Core Failure Mode

RAG pipelines fail silently at ingestion. The failure is invisible because the vector database is not a quality gate -- it is a storage layer that accepts any text you give it. Feed it a corrupted OCR chunk, a repeated fragment, or an encoding-garbled paragraph, and it will index it faithfully. The embedding will look fine. The retrieval similarity score will look fine. The problem only manifests downstream, when the model hallucinates based on a corrupted context or fails to retrieve because the relevant chunk was OCR-garbled into unrecognizable tokens.

The fundamental issue: vector databases accept any text you give them. They do not know that a chunk is a repeated fragment, garbled OCR output, or encoding-corrupted paragraph. DocIngestQA moves the quality gate to before indexing, where it is cheap to fix.

1.2 Concrete Examples of Bad Data Getting Through

OCR Garble: A scanned PDF of a financial report gets extracted with `non_alpha_ratio = 0.48`. Characters like 'Ã(c)', 'a€(TM)', 'l1l1', '0000' fill the chunk. The embedding model tokenizes this as noise. When a user asks 'What was Q3 revenue?' the retrieval returns the garbled chunk -- and the model either hallucinates a number or says 'I don't know', with no trace to the actual cause.

Missing Pages: A 45-page annual report is chunked but pages 20-28 (the risk factors section) fail to extract due to a scanned image embed in the PDF. The index silently omits 9 pages. When risk factor questions are asked, retrieval returns nothing. The user attributes this to the model -- not the missing pages.

Duplicate Chunks: A chunker with sliding-window overlap produces the same 200-word passage in 3 consecutive chunks. All 3 are indexed. When this passage is retrieved, the model's context window contains three copies of the same text, consuming space that could be used for other relevant content. Retrieval precision drops.

Encoding Corruption (Mojibake): A PDF using Windows-1252 encoding is extracted as UTF-8 without proper transcoding. Apostrophes become 'ae_tm', em-dashes become 'ae_dsh'. These are not just cosmetic -- the embedding vector for 'donâ€™(TM)t' is completely different from 'don\'t'. Semantic search breaks for any query that would have matched the original text.

Split at Wrong Boundary: A recursive chunker splits at 512 tokens regardless of sentence boundaries. A chunk ends: 'The maximum allowable loan-to-value ratio for commercial properties in metropolitan zones is'. The conclusion is in the next chunk. When retrieved, neither chunk answers the question correctly -- the retrieved chunk is literally incomplete.

1.3 Why This Is Hard to Catch Without a Systematic Tool

Manual inspection of chunk JSONL files is impractical at scale. A corpus of 150K chunks cannot be visually reviewed. Generic data quality tools (Great Expectations, dbt tests) check schema structure but have no

concept of what makes a RAG chunk semantically useful. LLM-based evaluation is circular: you cannot use the model that will read the chunks to evaluate whether the chunks are good for that model. And the failures are domain-specific: OCR noise patterns, page gap detection, split boundary heuristics, and encoding corruption detection all require understanding of the RAG ingestion context.

1.4 What DocIngestQA Actually Does About It

DocIngestQA runs 11 deterministic checks against exported chunk JSONL files, produces PASS/WARN/FAIL status at the check and corpus level, and generates a structured report with chunk-level evidence. It gates the ingestion pipeline: FAIL status stops indexing, WARN status flags issues for review, PASS proceeds to the vector store. The checks are cheap (no LLM calls, no model inference), reproducible (deterministic given the same input), and deployable in CI pipelines.

Section 2 -- Architecture Deep Dive: All 11 Checks

2.1 System Architecture

```
Raw Documents (PDF, HTML, DOCX, scanned images) | v Chunking Pipeline (LangChain / LlamaIndex / custom splitter) | v -- exports chunks.jsonl + optional source_manifest.json
v [DocIngestQA Audit -- 11 deterministic checks] <-- THIS TOOL | FAIL --> Stop. Fix extraction before indexing. WARN --> Review findings. Proceed with documented exceptions. PASS --> Proceed. | v Vector Store Indexing (Chroma, Pinecone, Weaviate, pgvector) | v RAG Application
```

2.2 v0.1 Checks (Checks 1-8)

Check	Status	What It Catches
input_summary	FAIL/PASS	Validates JSONL structure: required fields present (chunk_id, source, page, text), no null IDs, no empty text. FAIL if no chunks loaded.
metadata_completeness	FAIL/WARN/PASS	Checks optional fields (section, doc_type, language, timestamp) for completeness. FAIL if >20% missing source/page, WARN if >5%.
document_coverage	FAIL/WARN/PASS	If manifest provided: every expected document has at least one chunk. FAIL on missing docs, WARN on orphan sources (chunks from unexpected sources).
page_coverage	FAIL/WARN/PASS	For each manifest document with expected page counts, verifies contiguous page coverage. FAIL if any pages missing, WARN if extra pages detected.
chunk_length	FAIL/WARN/PASS	Flags chunks below min_chunk_chars (80) or above max_chunk_chars (1500). FAIL if >10% empty, WARN on short/long outliers.
ocr_noise	FAIL/WARN/PASS	Scans for U+FFFD replacement chars, repeated junk character runs, non-printable character ratio. FAIL if >20% noisy, WARN on partial signals.
duplicate_chunks	FAIL/WARN/PASS	SHA-1 hash for exact duplicates; Jaccard similarity on 5-grams for near-duplicates (threshold 0.85). FAIL if ≥30 duplicate pairs.
source_distribution	WARN/PASS	Flags if any single source contributes >80% of all chunks -- corpus bias that makes retrieval source-dependent regardless of query.

2.3 v0.2 Checks (Checks 9-11)

Check	Status	What It Catches
chunk_overlap	FAIL/WARN/PASS	Jaccard similarity on 4-grams between consecutive chunks from the same source. FAIL if ≥ 15 high-overlap (≥ 0.70) pairs, WARN if ≥ 3 moderate-overlap (≥ 0.40) pairs.
encoding_health	FAIL/WARN/PASS	Detects null bytes, BOM markers, control characters, mojibake sequences (Latin-1/UTF-8 mismatch). FAIL on null bytes or mojibake rate $\geq 20\%$.
split_quality	FAIL/WARN/PASS	Detects mid-sentence starts, mid-sentence ends, navigation fragments (bare page numbers, TOC entries). FAIL if $\geq 30\%$ of chunks flagged.

2.4 Severity Hierarchy

The four-level status hierarchy is: FAIL > WARN > INSUFFICIENT_INPUT > PASS. FAIL indicates structural issues that will definitively degrade retrieval. WARN indicates quality issues that reduce retrieval quality but don't break the index. INSUFFICIENT_INPUT means the check could not run due to missing configuration (e.g. no manifest). This is explicitly NOT PASS -- it must be documented. The overall corpus status is the maximum severity across all checks.

2.5 The AuditConfig Thresholds

Parameter	Default	Controls
min_chunk_chars	80	Short chunk threshold (below = fragment)
max_chunk_chars	1500	Long chunk threshold (above = context overflow risk)
noisy_text_ratio_threshold	0.05	Non-printable character rate that triggers OCR WARN
replacement_char_threshold	3	U+FFFD count per chunk before OCR flag
near_duplicate_jaccard_threshold	0.85	5-gram Jaccard for near-duplicate detection
fail_duplicate_pair_count	30	Duplicate pair count that triggers FAIL
overlap_jaccard_warn_threshold	0.40	4-gram Jaccard moderate overlap (WARN)
overlap_jaccard_fail_threshold	0.70	4-gram Jaccard high overlap (FAIL)

mojibake_fail_rate	0.20	Mojibake chunk rate that triggers encoding FAIL
fail_bad_split_rate	0.30	Bad split boundary rate that triggers split FAIL

Section 3 -- Algorithm Explanations: Math and Worked Examples

3.1 SHA-1 Hashing for Exact Duplicate Detection

Exact duplicate detection uses SHA-1 hashes of normalized chunk text. Normalization applies two transforms: lowercase the text, and collapse all whitespace runs to a single space. The normalized text is then SHA-1 hashed to a 40-character hex digest.

```
normalize(text) = collapse_whitespace(text.lower()) hash = SHA-1(normalize(text))
```

Two chunks with the same hash after normalization are exact duplicates regardless of original spacing or casing. This is deterministic and $O(n)$ in the number of chunks. The trade-off: near-duplicates differing in a single word, a punctuation mark, or a sentence insertion are NOT caught by this check. That is the job of Jaccard similarity on n-grams, covered in 3.2.

3.2 Jaccard Similarity on N-grams for Near-Duplicate Detection

Near-duplicate detection uses Jaccard similarity on 5-grams (sets of 5 consecutive tokens). For two chunks A and B:

$$J(A, B) = |\text{ngrams}(A) \cap \text{ngrams}(B)| / |\text{ngrams}(A) \cup \text{ngrams}(B)|$$

Worked example with two near-identical chunks:

```
Chunk A: "The loan was approved on January 12th subject to income verification and credit check"
Chunk B: "The loan was approved on January 12th subject to income verification and final credit check"
5-grams of A: {(the, loan, was, approved, on), (loan, was, approved, on, january), ..., (income, verification, and, credit, check)}
5-grams of B: {(the, loan, was, approved, on), ..., (income, verification, and, final, credit)}
Shared 5-grams: ~14 of 16 total => J(A,B) ≈ 0.875 > threshold(0.85) => NEAR DUPLICATE flagged
```

Why 5-grams and not 3-grams? 3-gram overlap on common phrases (e.g. 'the', 'of the', 'in the context') would produce false positives for topically similar but distinct chunks. 5-grams are long enough to require actual shared content structure.

3.3 Chunk Overlap via 4-gram Jaccard (Check 9)

The chunk_overlap check is similar but targets consecutive pairs from the same source, using 4-grams (shorter) and different thresholds. Consecutive chunk overlap is expected in sliding-window splitters -- the check distinguishes healthy overlap (10-30%) from wasteful over-overlap (>40%) that means essentially the same passage is indexed twice.

Worked example -- sliding window splitter with step too small:

```
Chunk[i]: "revenue grew 34 percent in Q3 driven by enterprise contract renewals"
Chunk[i+1]: "grew 34 percent in Q3 driven by enterprise contract renewals exceeding"
4-gram overlap = {(grew, 34, percent, in), (34, percent, in, q3), ...} = 7/10 = 0.70 Verdict:
HIGH overlap pair -- FAIL threshold (≥ 0.70)
```

3.4 OCR Noise Detection -- Non-Printable Character Ratio

For each chunk, the check computes the fraction of characters that are not printable ASCII or common Unicode letters. Three independent signals are checked:

```
1. replacement_char_count: count of U+FFFD (Unicode replacement char) -- a direct signal that the extraction library encountered un-decodable bytes. 2. printable_noise_ratio: fraction of characters that are not printable ASCII -- high in scanned PDFs with failed OCR. 3. has_repeated_char_run: regex search for 5+ consecutive identical characters (e.g. 'aaaaaaaa', '11111') -- a common OCR misread pattern for image artifacts.
```

3.5 Mojibake Detection -- Encoding Health (Check 10)

Mojibake (Japanese for 'character transformation') occurs when bytes encoded in one charset are decoded using a different charset. Most common: Windows-1252 or Latin-1 files decoded as UTF-8. The byte sequence for an em dash (0x96 in Windows-1252) becomes the two-byte sequence 0xC2 0x96 in UTF-8, which renders as 'Âx96'. Apostrophes (0x92) become 'a€(TM)'.

Detection: the `mojibake_count()` function searches for a curated regex of known mojibake sequences: [mojibake patterns: Atilde+e, a-euro-tm, Acirc, etc.]. These are the specific byte patterns that result from the most common encoding mismatches in PDF extraction pipelines.

3.6 Split Quality -- Sentence Boundary Detection (Check 11)

Three independent signals for poor split quality:

```
starts_mid_sentence: the first character of the chunk is lowercase AND is not a proper noun or number -- indicating the chunk starts mid-sentence. ends_mid_sentence: the last non-whitespace character of the chunk is not a sentence terminal (. ! ? " ] ) AND the next chunk (if available) does not start with a capital -- indicating the split cut through a sentence. is_navigation_fragment: the chunk text is short (<40 chars) and matches patterns for page numbers, TOC entries, or section headers with no body content.
```

Section 4 -- Design Decisions: Why Things Are the Way They Are

Why Deterministic Checks Only -- No LLM Calls

A quality gate that requires LLM calls to evaluate is (1) expensive at scale -- you cannot afford to embed every chunk twice, (2) non-reproducible -- LLM outputs vary by temperature and model version, making the gate non-deterministic, and (3) circular -- using an LLM to evaluate chunks that will be used for LLM retrieval creates a dependency loop. Deterministic checks are cheap, reproducible, and produce the same result every time for the same input. This matters for CI pipelines where consistency is required.

Why These 11 Checks and Not Others

The 11 checks were selected by mapping every known RAG ingestion failure mode to a detectable signal. v0.1 focused on the most common structural issues: missing fields, missing documents, missing pages, length violations, OCR noise, duplicates, and distribution imbalance. v0.2 added the three failure modes that require pairwise comparison: chunk overlap (requiring consecutive chunk comparison), encoding health (requiring character-level analysis), and split quality (requiring boundary pattern matching). Every check that did not have a deterministic signal with a defensible threshold was deferred to FUTURE.

Why WARN vs FAIL -- The Severity Distinction

FAIL is reserved for issues that are structural impossibilities for a useful RAG system: a corpus with no chunks (nothing to index), missing expected documents (retrieval blind spots), or encoding corruption above threshold (semantic destruction). WARN is for issues where the corpus is still indexable but retrieval quality is degraded -- short chunks, page gaps, OCR noise, duplicates. The distinction matters for CI pipeline design: FAIL blocks, WARN requires documented override.

Why INSUFFICIENT_INPUT is Not PASS

Document coverage, page coverage, and temporal checks require configuration that may not be available: a source manifest for document/page coverage, timestamp columns for temporal checks. When that configuration is absent, the check reports INSUFFICIENT_INPUT rather than PASS. Treating it as PASS would hide the fact that the check was never run. INSUFFICIENT_INPUT appears explicitly in the report so downstream consumers know the check was skipped, not passed.

Why PyPI Published

PyPI publication means any RAG team can run `pip install docingestqa` without cloning the repository. The zero-external-dependency design (beyond standard library) means it installs without conflicts in any ML environment. The CLI interface makes it usable in CI without writing Python code.

Why the Interpretation Note is Mandatory

DocIngestQA checks ingestion quality only. It does not assess retrieval relevance, answer faithfulness, semantic coherence, or chunk informativeness. A corpus that passes all 11 checks is free from structural defects -- it is not guaranteed to produce good retrieval results. This note is mandatory on every output format and cannot be suppressed. The reason: reports shared with non-technical stakeholders have historically been interpreted as 'RAG system is correct', which is incorrect.

Section 5 -- Honest Positioning: What Is and Is Not Built

Truth is a competitive advantage in interviews. Being precise about what is implemented, what is simulated, and what is planned demonstrates engineering judgment, not limitation.

[IMPLEMENTED]	All 11 checks run deterministically on real JSONL input
[IMPLEMENTED]	PASS/WARN/FAIL/INSUFFICIENT_INPUT status hierarchy
[IMPLEMENTED]	JSON, Markdown, HTML output formats with chunk-level evidence
[IMPLEMENTED]	CLI (python -m docingestqa audit) with exit codes 0/1/2/3
[IMPLEMENTED]	PyPI published (pip install docingestqa)
[IMPLEMENTED]	27/27 tests passing
[IMPLEMENTED]	Mandatory interpretation note on all output formats
[IMPLEMENTED]	AuditConfig with configurable thresholds for all checks
[IMPLEMENTED]	SHA-1 exact duplicate + 5-gram Jaccard near-duplicate detection
[IMPLEMENTED]	Mojibake detection with curated regex of known sequences
[IMPLEMENTED]	4-gram Jaccard overlap for consecutive chunk pairs
[IMPLEMENTED]	Sentence boundary heuristics for split quality
[SIMULATED]	150-chunk demo corpus with deliberately injected defects
[SIMULATED]	Demo fix workflow (fix_demo_issues.py showing remediation steps)
[FUTURE]	Semantic coherence check using embedding cosine similarity within chunks
[FUTURE]	LLM-narrated summary option for non-technical stakeholders
[FUTURE]	Per-source report sections for multi-document corpora
[FUTURE]	Configurable severity overrides (promote WARN to FAIL for specific checks)
[FUTURE]	Semantic duplicate detection (embedding similarity, not n-gram overlap)
[FUTURE]	Integration with LangSmith or RAGAS for end-to-end pipeline audit

Section 6 -- Q&A; Battery (30+ Pairs)

These questions are the ones FAANG interviewers actually ask about auditing tools. Every answer explains WHY, not just WHAT. Memorize the reasoning, not the words.

Q1: Walk me through what happens when you run `audit_corpus()` on a 1 GB PDF corpus

Before `audit_corpus()` runs, your pipeline has already extracted the PDFs and exported a `chunks.jsonl` file -- `DocIngestQA` does not parse PDFs itself. You call `IngestionAuditor(chunks_path='chunks.jsonl', documents_path='manifest.json', config=AuditConfig())`. The auditor loads the JSONL into a list of `Chunk` objects (each with `chunk_id`, `source`, `page`, `text` and optional `metadata`). It then runs all 11 checks sequentially. Checks 1-8 are $O(n)$: each iterates the chunk list once. Check 9 (overlap) is $O(n)$ within each source group -- it sorts chunks by page and compares consecutive pairs. Check 10 (encoding) is $O(n)$. Check 11 (split quality) is $O(n)$. Total complexity is $O(n)$ in chunk count. For a 1 GB corpus that produces ~150K chunks, each check runs in seconds on modern hardware. The bottleneck is JSONL loading, not check execution. The output is an `IngestionAuditReport` with per-check `CheckResult` objects containing status, summary, metrics, and a list of `Issue` objects with chunk-level evidence. You call `report.to_html('audit.html')` to produce the self-contained review report.

Q2: What's the difference between a corrupt chunk and an incomplete chunk?

A corrupt chunk has damaged content -- encoding corruption (mojibake), OCR garble (non-printable characters, U+FFFD replacement chars), or null bytes. The text exists but is semantically destroyed. The embedding this chunk produces is meaningless. An incomplete chunk has valid content but is missing context -- it starts or ends mid-sentence, or it was part of a longer passage that was split incorrectly. The text is readable, but as a retrieval unit it is insufficient: the answer to a query may span the boundary of where this chunk was cut. `DocIngestQA` detects corrupt chunks primarily via Check 10 (`encoding_health`) and Check 6 (`ocr_noise`). Incomplete chunks are detected primarily via Check 11 (`split_quality`). The remediation is also different: corrupt chunks require re-extraction with proper encoding; incomplete chunks require re-chunking with different boundary rules.

Q3: Why 11 checks? How did you decide which to include?

The 11 checks map directly to the known failure modes of RAG document ingestion pipelines. The selection criterion was: does this failure mode have a deterministic signal that can be computed cheaply on the chunk text or metadata alone? Input structure (1) and metadata completeness (2) catch pipeline export bugs. Document coverage (3) and page coverage (4) catch chunker failures on specific documents. Chunk length (5) catches configuration errors. OCR noise (6) catches scan quality issues. Duplicate detection (7) catches overlap configuration and source duplication. Source distribution (8) catches corpus bias. Chunk overlap (9) catches splitter step misconfiguration. Encoding health (10) catches extraction encoding bugs. Split quality (11) catches boundary rule misconfiguration. Each check targets a specific failure mode with a specific, defensible detection method. Checks without defensible deterministic signals -- semantic coherence, answer faithfulness -- were explicitly deferred to FUTURE.

Q4: What does your overlap detector actually compute -- is it set intersection or something else?

It computes Jaccard similarity on 4-gram token sets. This is set intersection divided by set union -- $J(A,B) = |A \cap B| / |A \cup B|$. The 4-grams are extracted from whitespace-tokenized text. A 4-gram is a tuple of 4 consecutive tokens: for 'revenue grew in Q3' the 4-grams are {(revenue, grew, in, Q3)}. The choice of 4-grams (not token-level overlap) is deliberate: token-level overlap would flag chunks that share common words without sharing content structure. 4-gram overlap requires that the same 4 consecutive words appear in both chunks -- a strong signal of content repetition. The WARN threshold is 0.40 (40% of 4-grams shared) and the FAIL threshold is 0.70 (70% shared). These thresholds are configurable via AuditConfig.

Q5: How does SHA-1 deduplication handle near-duplicates?

It doesn't -- that's intentional and correctly partitioned. SHA-1 is used only for exact duplicates: two chunks with identical normalized text (after lowercasing and whitespace collapse) produce the same hash and are flagged as exact duplicates. Near-duplicates -- chunks that differ in a word, a punctuation mark, or a sentence -- produce completely different hashes. SHA-1 was not designed for approximate matching and should not be used for it. Near-duplicate detection uses a separate algorithm: 5-gram Jaccard similarity with a threshold of 0.85. This separation is a deliberate design choice: exact duplicate detection is $O(n)$ via hashing; near-duplicate detection is $O(n^2)$ in the worst case but is bounded by thresholding. Mixing them would create a single slow algorithm that does both imprecisely.

Q6: What's your false positive rate on the OCR noise check?

The OCR noise check has three independent signals, each with different false positive profiles. Replacement character (U+FFFD) detection has near-zero false positive rate -- U+FFFD only appears when a decoder encounters an un-decodable byte. It is not a normal text character. Non-printable ratio detection has a higher false positive rate on: mathematical PDFs (LaTeX symbols), code chunks (backtick formatting), and documents with Unicode symbols. The threshold of 0.05 (5% non-printable) was calibrated against real PDF corpora and may need adjustment for technical documentation. Repeated character run detection (regex for 5+ identical chars) can false-positive on emphasis patterns ('-----' dividers, 'NOTE:' formatting). In practice, the check is designed to over-report rather than under-report -- the cost of a false positive is a human reviewing a flagged chunk; the cost of a false negative is an OCR-garbled chunk making it into production.

Q7: How long does auditing 10K chunks take vs 150K chunks?

DocIngestQA's checks are $O(n)$ in chunk count, so runtime scales linearly. For 10K chunks: roughly 2-5 seconds on a modern laptop (2022 M2 MacBook). For 150K chunks: roughly 30-75 seconds. The dominant cost is JSONL loading and the pairwise overlap check (Check 9), which iterates consecutive pairs within each source. The encoding health check (Check 10) is the second most expensive: it runs regex matches on every chunk's text. Memory usage scales with chunk count: 150K chunks at ~500 chars average is roughly 75 MB of text in memory -- negligible for any modern process. The CLI mode reads the JSONL in a streaming pass before check execution, so the peak memory is one full pass through the chunk list. At 1M+ chunks, streaming would need to replace the in-memory list -- this is a FUTURE item.

Q8: What happens if the manifest JSONL is malformed?

The manifest is a JSON array (not JSONL -- it's `source_manifest.json`, not a line-delimited file). If the manifest is malformed JSON, the `readers.py load_manifest()` function raises a `JSONDecodeError` which propagates up to the `IngestionAuditor` initialization. The checks that depend on the manifest (`document_coverage` and `page_coverage`) will not run. If the manifest is valid JSON but has missing fields (e.g. a source object without the 'pages' key), `page_coverage` silently skips that document (it only checks documents where pages is provided). If the manifest is valid but empty, `document_coverage` returns PASS (no expected documents, nothing missing). This is a known limitation: malformed manifests do not produce a graceful `INSUFFICIENT_INPUT` -- they raise exceptions. A v0.3 improvement would be manifest validation that returns `INSUFFICIENT_INPUT` with a clear error message rather than raising.

Q9: How would you extend this to detect semantic duplicates vs exact duplicates?

Semantic duplicate detection requires embedding similarity, which is fundamentally different from the current approach. The design would be: (1) embed all chunks using a sentence transformer (e.g. `all-MiniLM-L6-v2`), (2) build an approximate nearest-neighbor index (FAISS flat or HNSW), (3) for each chunk, find the k nearest neighbors above a cosine similarity threshold (e.g. 0.92), (4) flag pairs as semantic duplicates if similarity > threshold AND the chunks are from different sources or non-consecutive positions. The challenge is that this changes the tool's character from deterministic to model-dependent: different embedding models produce different duplicate sets. The threshold also requires calibration -- 0.92 cosine similarity means similar but not identical content. This would be a FUTURE check, clearly labeled as model-dependent, with the embedding model version pinned in the report so results are reproducible across runs using the same model.

Q10: What's the difference between what DocIngestQA does and what LlamaIndex's built-in validation does?

LlamaIndex's built-in node processing does schema validation (required fields present) and basic metadata population, but it does not run quality checks on the content of chunks. LlamaIndex validates that your Node objects are correctly structured for its internal processing pipeline -- it does not check whether the text content is OCR-corrupted, whether pages are missing, whether chunks overlap excessively, or whether encoding corruption has destroyed semantic content. DocIngestQA operates on the exported JSONL after the chunking step, checking the quality of the content itself. The two tools are complementary: LlamaIndex ensures structural correctness for its own pipeline; DocIngestQA ensures content quality for any pipeline's output. You can run DocIngestQA on chunks produced by LlamaIndex, LangChain, or any custom splitter.

Q11: What are the 3 checks that can produce FAIL and why those specifically?

Check 1 (input_summary): FAIL if no chunks are loaded. A corpus with zero chunks is not auditable and not indexable. This is the most fundamental structural violation. Check 4 (page_coverage): FAIL if any expected pages are missing from the chunk set. Missing pages represent retrieval blind spots -- queries about content on those pages will never be answered correctly. This is a structural gap, not a quality issue. Check 10 (encoding_health): FAIL if null bytes are present OR mojibake rate $\geq 20\%$. Encoding corruption destroys semantic content. A chunk with mojibake text produces a completely wrong embedding. At 20% corpus corruption, the retrieval system is fundamentally compromised. The 8 WARN checks cover issues where the corpus is still indexable but retrieval quality is degraded. The FAIL/WARN distinction reflects whether the issue is a structural impossibility or a quality degradation.

Q12: How does DocIngestQA connect to GoldenSetAuditor?

DocIngestQA and GoldenSetAuditor are complementary quality gates for different parts of the RAG evaluation pipeline. DocIngestQA audits the source corpus -- are the chunks structurally sound before indexing? GoldenSetAuditor audits the evaluation dataset -- are the reference question-answer pairs trustworthy before benchmarking? Both must pass for a RAG system's evaluation metrics to be meaningful. Here's why: if the source corpus has OCR-corrupted chunks (DocIngestQA would catch this), the retrieval metrics reflect chunk quality, not model quality. If the golden set has conflicting labels or unanswerable questions (GoldenSetAuditor would catch this), the evaluation metrics are unreliable regardless of how good the retrieval is. A system with clean chunks but conflicting golden labels produces unreliable benchmark scores. A system with clean golden labels but OCR-corrupted chunks produces low retrieval metrics that look like a model problem. You need both gates.

Q13: What would you add in v2? Give 5 concrete improvements.

1. Semantic coherence check: embed each chunk and compute the average pairwise cosine similarity between a chunk and its surrounding chunks in the same source. A chunk that is semantically dissimilar from its neighbors (cosine < 0.3) is likely a mis-extracted fragment or a table/figure that was included in a text chunk. This requires choosing and pinning an embedding model.
2. Graceful manifest error handling: replace JSONDecodeError propagation with INSUFFICIENT_INPUT status and a clear error message.
3. Per-source HTML report sections: the current HTML report is corpus-level. v2 would break findings down by source document, making it easier for document owners to review issues in their specific files.
4. Configurable severity overrides: allow AuditConfig to promote specific WARN checks to FAIL for stricter pipelines (e.g. FAIL on any OCR noise for medical documents).
5. Streaming mode for 1M+ chunk corpora: replace in-memory list loading with a streaming pass that maintains running statistics for each check without holding all chunks in memory.

Q14: How does the source_distribution check work and when does it matter most?

The source_distribution check computes the fraction of total chunks contributed by each source document: $\text{chunks_from_source} / \text{total_chunks}$. If any single source exceeds max_source_fraction (default 80%), it's flagged WARN. This matters most in multi-document RAG systems where the corpus is supposed to give equal retrieval opportunity to all source documents. If source A contributes 85% of chunks and sources B-F contribute 3% each, then for any query, the top-k retrieved chunks will overwhelmingly come from source A -- not because source A is more relevant, but because it dominates the vector space. This produces retrieval bias that looks like correct behavior until you notice that answers always cite the same document. The check is WARN-only (not FAIL) because corpus imbalance is sometimes intentional: a support knowledge base might legitimately have one large product manual and several small supplementary guides.

Q15: What is the source manifest and what happens without one?

The source manifest (source_manifest.json) is a JSON array where each entry specifies a source document name and expected page count: [{source: doc.pdf, pages: 45}, ...]. It enables the two coverage checks: document_coverage (every expected document has at least one chunk) and page_coverage (every expected page has at least one chunk). Without a manifest, both checks return INSUFFICIENT_INPUT -- they cannot run because there is no ground truth to compare against. The auditor does not fail or skip -- it explicitly reports INSUFFICIENT_INPUT so the report consumer knows these checks were not evaluated. This distinction is important: a corpus that passes 9 checks but has 2 INSUFFICIENT_INPUT checks has a different risk profile than a corpus with 11 PASS checks.

Q16: Walk me through the encoding_health check in detail

The check iterates every chunk and runs four independent detectors on the text: (1) null_byte_count(): counts occurrences of '\x00' -- null bytes corrupt any text processing and are always extraction artifacts. (2) bom_detected(): looks for BOM markers (\xef\xbb\xbf, \xff\xfe, \xfe\xff) at the start of text -- these are encoding markers that should be stripped during extraction but sometimes aren't. (3) mojibake_count(): applies a regex with known mojibake sequences (Ã(c), a€(TM), etc.) to count encoding corruption artifacts. (4) control_char_count(): counts characters in Unicode 'control' category (Cc) excluding standard whitespace. Status: FAIL if any null bytes present (null_byte_fail_threshold=1) OR if mojibake rate $\geq$ mojibake_fail_rate (20%). WARN if mojibake rate $\geq$ 5% OR BOM present OR any control characters found.

Q17: How does the CLI work and how would you use it in a GitHub Actions CI pipeline?

The CLI is invoked as `python -m docingestqa audit chunks.jsonl --manifest source_manifest.json --out outputs/`. It exits with code 0 (PASS), 1 (WARN), 2 (FAIL), or 3 (INSUFFICIENT_INPUT). In a GitHub Actions CI pipeline: `steps:\n- name: Audit chunks\nrun: |\n pip install docingestqa\n python -m docingestqa audit outputs/chunks.jsonl --manifest source_manifest.json\n# Exit code 2 (FAIL) will fail the step and block the workflow\nThe WARN exit code (1) does not fail the step by default -- you can add || [$? -eq 1] to the run command to treat WARNs as acceptable, or use if: failure() to post the audit report as a PR comment even on WARN.`

Q18: Can DocIngestQA run on chunks from any RAG framework, not just LangChain?

Yes, completely framework-agnostic. DocIngestQA operates on JSONL files where each line is a chunk with fields: `chunk_id`, `source`, `page`, `text` (optional: `section`, `doc_type`, `language`, `timestamp`). Any chunking framework -- LangChain, LlamaIndex, Haystack, Unstructured, custom splitters -- can export to this format. If your framework doesn't export JSONL natively, the conversion is trivial: iterate your chunks, extract the required fields, write one JSON object per line. The schema is intentionally minimal -- `chunk_id` and `text` are the only truly required fields. Everything else enriches the audit but is not mandatory.

Q19: What's the difference between `page_coverage FAIL` and `page_coverage WARN`?

`page_coverage FAIL` means expected pages are missing from the chunk set -- pages that should have chunks according to the manifest have no chunks at all. This is a retrieval blind spot: content on those pages is unindexed. `page_coverage WARN` means page numbers in the chunk set exceed the expected page count according to the manifest -- there are chunks with page numbers higher than the document's declared page count. This suggests either the manifest is wrong (declared 45 pages but the PDF is actually 52 pages) or the extractor is generating phantom page numbers. Missing pages are FAILs because they definitively cause retrieval blind spots. Extra pages are WARNs because they may be legitimate (manifest misconfigured) and don't necessarily cause retrieval failures -- they add content, not remove it.

Q20: How does DocIngestQA handle multi-language corpora?

The language field in chunk metadata is validated for completeness in Check 2 (`metadata_completeness`). OCR noise detection in Check 6 uses character class patterns that are primarily Latin-alphabet based. For CJK corpora, the non-alpha ratio check would produce false positives because CJK characters are not ASCII letters. In the current implementation, the language metadata field is collected but the checks are not yet language-aware -- this is a known limitation. The recommended practice for CJK corpora is to set `noisy_text_ratio_threshold` to a higher value (0.40+) in `AuditConfig`, effectively disabling the false-positive-prone signal. Sentence boundary detection in Check 11 uses terminal punctuation patterns that are language-specific; non-Latin corpora should set the `fail_bad_split_rate` threshold higher to avoid false positives on valid non-Latin sentence structures.

Q21: What's the difference between DocIngestQA and a simple Python lint script?

A lint script checks syntax. DocIngestQA checks semantics. The chunks are syntactically valid JSON with all required fields present -- that's what Check 1 covers. But a corpus that passes Check 1 can still have: pages 20-35 missing from a 50-page document (page coverage gap), 8 chunks with OCR-garbled text that tokenizes as noise (OCR noise), 4 duplicate chunk pairs that waste index capacity (duplicate detection), 3 consecutive chunks with 70% content overlap (overlap check). None of these are syntax problems. All of them degrade RAG retrieval quality. The semantic knowledge of what makes a good RAG chunk -- expected page contiguity, safe non-alpha character ratios, reasonable duplicate thresholds, healthy split boundaries -- is what DocIngestQA packages into systematic checks. A lint script doesn't have domain knowledge. DocIngestQA does.

Q22: How does the near-duplicate check avoid $O(n^2)$ complexity?

The naive near-duplicate detection algorithm compares every pair of chunks, which is $O(n^2)$. For 150K chunks that would be 11.25 billion comparisons -- intractable. DocIngestQA's current implementation uses a hash-bucketing approach: 1. All chunks are SHA-1 hashed (exact duplicates are found in a single pass). 2. For near-duplicate detection, chunks are sorted by source and grouped. 3. Near-duplicate Jaccard comparison is run only within source groups and between consecutive chunks in the page-ordered sequence. This is a practical heuristic that catches the most common near-duplicate pattern (sliding-window splitter producing overlapping consecutive chunks) without full $O(n^2)$ comparison. It does miss non-consecutive near-duplicates (the same passage appearing in non-adjacent chunks from different pages). Full near-duplicate detection at scale would require LSH (Locality-Sensitive Hashing) on MinHash signatures -- a FUTURE item.

Q23: What's the most common FAIL finding in production corpora?

Encoding corruption (Check 10) -- specifically mojibake from PDF extraction. PDFs that use non-standard font encoding, CJK character sets, or ligature substitutions frequently produce mojibake when extracted with common libraries (pdfplumber, PyMuPDF). The pattern is systematic: a specific document type or extraction configuration affects all chunks from those sources. The second most common FAIL is page coverage (Check 4): scanned PDFs with image-only pages that the extractor cannot OCR produce coverage gaps. A 50-page annual report where pages 20-28 are scanned charts will have no chunks for those pages. Input structure (Check 1) rarely FAILs in practice -- it only fires if the pipeline produces zero chunks, which is usually caught during development.

Q24: What is chunk_overlap and why is zero overlap also a problem?

Chunk overlap is the fraction of content shared between consecutive chunks from the same source. A sliding-window splitter with window=512 and step=400 produces 112-token overlap between consecutive chunks -- about 22%. Too much overlap (>40%, Jaccard threshold): the same passage appears in multiple chunks. When retrieved, the model's context window fills with near-identical content, crowding out potentially relevant different content. Index size is also wasted. Zero overlap (hard cuts): a sentence that spans the chunk boundary is split in two. Chunk[i] ends mid-sentence; Chunk[i+1] starts mid-sentence. Both chunks are semantically incomplete retrieval units. The split_quality check (Check 11) catches the boundary manifestation; the overlap check (Check 9) catches the configuration that produced it. The healthy range is 10-20% overlap -- enough for continuity, not so much that content is duplicated.

Q25: How would you test the 11 checks in isolation?

The test suite (27/27 passing) uses a fixture-based approach. For each check: (1) construct a minimal synthetic chunk set with exactly the defect the check targets, (2) assert the check produces the expected status (FAIL/WARN/PASS), (3) assert the Issue list contains the expected evidence fields. For example, `test_ocr_noise` constructs chunks with embedded U+FFFD characters and asserts `check_ocr_noise` returns WARN with the correct chunk IDs flagged. `test_encoding_health` constructs chunks with mojibake sequences (the actual `Ã(c)` string) and asserts FAIL when the rate exceeds `mojibake_fail_rate`. Negative tests verify that clean chunks produce PASS for each check. Integration tests run the full 11-check pipeline on the demo corpus and assert the expected overall status (WARN, because the demo corpus has deliberate defects).

Q26: Explain the status hierarchy -- what happens when multiple checks fire?

The overall corpus status is the maximum severity across all check results. The hierarchy is: FAIL > WARN > INSUFFICIENT_INPUT > PASS. If Check 10 (`encoding_health`) FAILs and Check 6 (`ocr_noise`) WARNs, the overall status is FAIL. If no checks FAIL but 5 checks WARN, the overall status is WARN. If all checks PASS except 2 return INSUFFICIENT_INPUT, the overall status is PASS (INSUFFICIENT_INPUT is below PASS in the hierarchy -- it does not raise the overall status). Wait -- this needs a clarification: INSUFFICIENT_INPUT checks are informational: they mean a check could not run, not that it found a problem. The overall status prioritizes actual findings (FAIL > WARN > PASS) and reports INSUFFICIENT_INPUT separately in the check-level results.

Q27: How does the tool output differ between JSON, Markdown, and HTML?

JSON: machine-readable structured output with the full schema -- `schema_version`, `metadata`, `executive_summary`, and a `checks` array where each check has its `CheckResult` with `status`, `summary`, `metrics`, `issues` (with chunk-level evidence), and `recommendation`. Intended for downstream processing: CI pipeline status checks, LLM-narrated summaries, feeding into dashboards. Markdown: human-readable check-by-check tables. Intended for PR comments and developer review. Each check's result is a row in a Markdown table with `status` and `summary`. HTML: self-contained single-file report with color-coded findings (FAIL in red, WARN in amber, PASS in green), chunk-level evidence (source, page, text preview). Intended for sharing with non-technical stakeholders -- document owners who don't read JSON.

Q28: If the audit shows WARN for split_quality on 15% of chunks, what would you do?

15% bad split rate is above the WARN threshold (10%) but below the FAIL threshold (30%). The recommended response: (1) Look at the evidence: which sources and pages have the most bad-split chunks? (2) Examine the actual split boundary examples -- are they consistently mid-sentence starts, mid-sentence ends, or navigation fragments? The pattern tells you which splitter configuration is wrong. (3) If mid-sentence endings dominate: reduce the chunk size or switch to a sentence-aware splitter that respects sentence terminals. (4) If navigation fragments dominate: add a pre-processing step that removes TOC entries, page headers, and footers before chunking. (5) Re-run extraction with the new configuration and re-audit. The goal is to get split_quality to PASS before indexing, but WARN is not a blocker -- you can proceed with documented awareness of the issue.

Q29: What's the relationship between chunk length and embedding quality?

Embedding models have context windows -- typically 128 to 512 tokens. A chunk that exceeds the embedding model's context window is truncated before embedding: the content beyond the window is silently dropped. This means a 1200-character chunk may have its most relevant sentence (at position 800) truncated out of the embedding. The retrieved chunk contains the answer, but the embedding doesn't encode it -- retrieval fails. This is the failure mode the max_chunk_chars threshold prevents. For very short chunks (<80 chars): single-sentence chunks often lack enough context for the model to generate a useful answer, even if retrieval succeeds. 'Revenue grew.' is a retrieved chunk, but it doesn't answer 'by how much' or 'why'. The min_chunk_chars threshold prevents these fragment chunks from entering the index.

Q30: What would change if you were auditing chunks destined for a BM25 index vs a vector store?

The core structural checks (1-5) are relevant for both: missing pages, metadata completeness, chunk length are quality issues regardless of the retrieval method. The OCR noise check (6) matters more for BM25: BM25 is term-frequency based, so OCR-garbled terms dilute the IDF statistics for correct terms. A chunk with 'Ã(c)' instead of 'e' is invisible to BM25 queries using the correct character. The encoding health check (10) is equally critical for both -- garbled text breaks tokenization in both systems. The overlap check (9) matters less for BM25 than for dense retrieval: BM25 scoring is term-independent, so repeated passages don't amplify the same passage to the top of results the same way cosine similarity does. A BM25-specific audit would add: vocabulary distribution check (one source dominating IDF statistics), stopword density check (chunks that are almost entirely stopwords), and term frequency outlier detection.

Q31: How would you extend DocIngestQA for multilingual corpora?

Three extensions needed: (1) Language-aware character set detection: the OCR noise check currently flags high non-ASCII ratio, which produces false positives for CJK, Arabic, Hebrew, and Cyrillic text. Extension: auto-detect chunk language using langdetect or fasttext, then skip the non-ASCII ratio check for non-Latin scripts. Use script-specific noise patterns instead (e.g. for CJK: detect interrupted character sequences, unusual code-point ranges). (2) Language-aware sentence boundary detection: the split_quality check uses Latin terminal punctuation patterns. For Arabic (right-to-left), CJK (period-based but different code point), and Hebrew, different boundary patterns apply. (3) Language consistency check: flag chunks within a source where the detected language differs from the declared language metadata -- a signal that the extractor incorrectly processed multilingual pages.

Q32: What does the demo corpus demonstrate and how is it structured?

The demo corpus (examples/assets/chunks.jsonl) contains 64 synthetic chunks across 10 source documents with deliberately seeded defects: - 2 chunks missing source or page metadata (metadata_completeness WARN), - 10 missing pages across 2 documents (page_coverage FAIL), - 3 short chunks below min_chunk_chars (chunk_length WARN), - 2 chunks with OCR noise signals (ocr_noise WARN), - 1 exact duplicate chunk pair (duplicate_chunks WARN), - 2 chunks with mojibake sequences (encoding_health WARN), - 4 chunks with poor split boundaries (split_quality WARN). Running audit_demo.py produces overall status FAIL (due to page_coverage FAIL) with 4 PASS, 6 WARN, 1 FAIL across 11 checks. This matches the README's documented output exactly -- the demo is a regression test as well as a demonstration.

Section 7 -- Failure Modes: What Breaks in the Auditor Itself

This section covers what DocIngestQA gets wrong -- not what it catches. Being able to articulate the limits of your own tool is the mark of an engineer who thought through the design rather than just built it.

False Negatives: What the Auditor Misses

Near-duplicates in non-consecutive positions: the overlap check (9) only compares consecutive chunks within a source. The same passage appearing at page 5 and page 42 of the same document -- or in two different documents -- is not caught. Semantic duplicates: two chunks that say the same thing in different words are not caught by any current check. Both the SHA-1 hash and the n-gram Jaccard will miss semantic equivalence. OCR noise in CJK or non-Latin corpora: the non-alpha ratio check produces false negatives (misses real OCR noise) in scripts where the check is disabled. Subtle encoding issues: some mojibake patterns are not in the curated regex list and will be missed. Split quality in specialized documents: table cells, mathematical expressions, and code blocks have different boundary expectations that the current heuristics don't handle.

False Positives: What the Auditor Flags Incorrectly

Mathematical/scientific documents: high non-alpha character ratio from LaTeX symbols, Greek letters, subscripts, and superscripts will trigger OCR noise WARNs on legitimate content. Code documentation: chunks containing code snippets with backticks, underscores, and symbol-heavy syntax will have elevated non-printable ratios. Deliberately short chunks: chapter headers, section titles, and structured metadata entries are intentionally short but serve a purpose. `min_chunk_chars` flags them. Near-duplicate false positives: two chunks from different parts of a legal document that use formulaic language ('pursuant to Section X of the Agreement, the parties agree that...') may hit the Jaccard threshold despite being legitimately different clauses.

Manifest-Dependent Checks Are Only as Good as the Manifest

If `source_manifest.json` incorrectly declares 45 pages for a document that actually has 52 pages, `page_coverage` will flag 7 'extra' pages as WARN when the extraction is actually correct. If the manifest is missing a document entirely, `document_coverage` PASS means 'all documents in the manifest are present' -- it does not mean 'all documents that should have been indexed are present.' The auditor cannot validate what it wasn't told to check for.

Threshold Miscalibration

All thresholds in `AuditConfig` have defaults calibrated for general English-language document corpora. These defaults may be wrong for: medical documents (high symbol density), legal documents (formulaic repetition triggers near-duplicate), technical documentation (code blocks trigger OCR noise), or non-English corpora. Teams using DocIngestQA in specialized domains should run the auditor on a known-good corpus first and adjust thresholds to minimize false positives before deploying it as a CI gate.

Streaming/Memory Limitations

The current implementation loads all chunks into memory as a list of Chunk objects. At 150K chunks this is manageable (~75 MB). At 1M+ chunks this becomes a memory issue -- loading 750 MB of text into a Python list will cause OOM on machines with limited RAM. The overlap check (which maintains per-source chunk lists) is particularly memory-intensive for sources with many chunks. Streaming mode is a FUTURE item.

Section 8 -- Scalability: Performance at 100K Chunks

8.1 Computational Complexity by Check

Check	Complexity	Bottleneck	100K Estimate
input_summary	$O(n)$	JSONL load	~0.5s
metadata_completeness	$O(n)$	Field iteration	~0.3s
document_coverage	$O(n)$	Set ops on sources	~0.1s
page_coverage	$O(n)$	Set ops per source	~0.2s
chunk_length	$O(n)$	len() per chunk	~0.2s
ocr_noise	$O(n \cdot \text{avg_len})$	Regex per chunk	~2s
duplicate_chunks	$O(n \cdot k)$	SHA-1 + n-gram tokenize	~5s
source_distribution	$O(n)$	Counter ops	~0.1s
chunk_overlap	$O(n \cdot k)$ per source	n-gram + Jaccard per pair	~8s
encoding_health	$O(n \cdot \text{avg_len})$	Regex per chunk	~3s
split_quality	$O(n)$	Boundary pattern match	~1s
TOTAL	$O(n)$	overlap + encoding	~20s

8.2 Memory Scaling

Corpus Size	Approx Chunks	Memory (List)	Recommendation
Small (100 docs, 50p)	~10K chunks	~5 MB	In-memory, no issue
Medium (500 docs, 50p)	~50K chunks	~25 MB	In-memory, no issue
Large (1000 docs, 150p)	~150K chunks	~75 MB	In-memory, monitor
Very Large (10K docs)	~1M+ chunks	~500 MB+	Use streaming mode (FUTURE)

8.3 Parallelism Opportunities

The 11 checks are currently run sequentially. Most checks are embarrassingly parallel -- each chunk is processed independently. The overlap check is the exception: it requires the sorted sequence of consecutive chunks within each source. A parallelized v0.3 could: (1) run per-chunk checks (ocr_noise, encoding_health, chunk_length, split_quality) in parallel using multiprocessing.Pool, (2) run the overlap check per-source in parallel (each source is independent). At 100K chunks, the current sequential runtime (~20s) is acceptable for a CI gate. At 1M+ chunks, parallelism would reduce runtime from ~3 minutes to ~30 seconds on a 6-core machine.

Section 9 -- Cross-Project Connections

9.1 The RAG Quality Stack

DocIngestQA is one layer in a three-layer RAG quality stack. Understanding the full stack shows architectural thinking, not just tool knowledge:

Layer	Tool	When It Runs	What It Guards
Pre-Indexing	DocIngestQA	After chunking, before vector store	Chunk structural quality
Pre-Evaluation	GoldenSetAuditor	After annotation, before benchmarking	Eval dataset quality
Pre-Training	FeatureLeakageLens	After feature eng, before model fit	Feature leakage

9.2 Connection to DevPulse (Version-Safe RAG Platform)

DevPulse is the portfolio's version-safe RAG platform. Its 9-chunk demo corpus passes through DocIngestQA before indexing. DevPulse achieves Recall@5 = 0.97 in its demo evaluation. That metric is meaningful only because the underlying chunks are structurally clean: no OCR noise, no page gaps, no encoding corruption. If you used DevPulse's conflict detector on OCR-corrupted chunks, the BM25 term statistics would be distorted (garbled OCR tokens get IDF scores), and the conflict detection precision would drop. DocIngestQA is what makes DevPulse's retrieval metrics trustworthy.

9.3 Connection to GoldenSetAuditor

GoldenSetAuditor audits evaluation Q&A; pairs before benchmarking. DocIngestQA audits the source corpus before indexing. You need both: - Clean corpus + clean golden set = reliable evaluation metrics. - Clean corpus + bad golden set = evaluation metrics that reflect annotation quality, not model quality. - Bad corpus + clean golden set = evaluation metrics that reflect chunk quality, not model quality. - Bad corpus + bad golden set = evaluation metrics that are meaningless. In the portfolio, these two tools are meant to be run sequentially as part of a RAG development workflow: DocIngestQA gates corpus ingestion; GoldenSetAuditor gates evaluation setup.

Section 10 -- Resume Claim Audit

Resume Claim (Exact Text):

"Built DocIngestQA, a pre-indexing QA auditor for RAG document ingestion pipelines that runs 11 deterministic checks on exported chunk JSONL files -- detecting missing pages, OCR noise, exact/near-duplicate chunks, mojibake encoding errors, sliding-window split overlap, and poor split boundaries -- with a CLI, configurable thresholds, structured JSON/Markdown/HTML output, and a mandatory truth boundary note that distinguishes ingestion quality signals from retrieval relevance or RAG correctness."

Claim	Status	Evidence
11 deterministic checks	[IMPLEMENTED]	Checks 1-8 (v0.1) and 9-11 (v0.2) all implemented and tested
Detecting missing pages	[IMPLEMENTED]	Check 4 (page_coverage): FAIL on any missing pages
OCR noise	[IMPLEMENTED]	Check 6 (ocr_noise): U+FFFD, non-printable ratio, repeated char runs
Exact/near-duplicate chunks	[IMPLEMENTED]	Check 7: SHA-1 for exact, 5-gram Jaccard for near-duplicates
Mojibake encoding errors	[IMPLEMENTED]	Check 10 (encoding_health): curated mojibake regex detection
Sliding-window split overlap	[IMPLEMENTED]	Check 9 (chunk_overlap): 4-gram Jaccard on consecutive pairs
Poor split boundaries	[IMPLEMENTED]	Check 11 (split_quality): mid-sentence and nav fragment detection
CLI	[IMPLEMENTED]	python -m docingestqa audit with exit codes 0/1/2/3
Configurable thresholds	[IMPLEMENTED]	AuditConfig with all thresholds documented
JSON/Markdown/HTML output	[IMPLEMENTED]	All three formats in report.py
Mandatory truth boundary	[IMPLEMENTED]	Cannot be suppressed from any output format

Section 11 -- Study Plan + Do Not Say This

11.1 30-Minute Pre-Interview Review

- Read checks.py from top to bottom -- know every check's logic by name
- Re-run examples/audit_demo.py and explain the FAIL finding (page_coverage)
- Be able to draw the pipeline architecture from memory (Raw Docs → Chunker → DocIngestQA → Vector Store)
- Know the 3 FAIL checks by heart: input_summary (empty corpus), page_coverage (missing pages), encoding_health (null bytes / mojibake rate)
- Be able to explain Jaccard similarity on n-grams with a concrete worked example
- Know the exit code contract: 0/1/2/3 and what each means for CI
- Know the difference between INSUFFICIENT_INPUT and PASS
- Be ready to explain the GoldenSetAuditor connection in 2 sentences

11.2 Do Not Say These Things

The following phrases will undermine your credibility. Avoid them.

DO NOT SAY: "DocIngestQA guarantees good RAG quality"

SAY INSTEAD: DocIngestQA checks ingestion quality only. Good chunks are necessary but not sufficient for good retrieval. Always add: 'it gates structural defects -- retrieval relevance is a separate concern.'

DO NOT SAY: "It catches all duplicates"

SAY INSTEAD: It catches exact duplicates (SHA-1) and near-duplicates in consecutive positions (Jaccard on n-grams). Non-consecutive near-duplicates and semantic duplicates are not caught. Be specific about the algorithm and its scope.

DO NOT SAY: "The false positive rate is zero"

SAY INSTEAD: The OCR noise check has false positives on mathematical, code, and non-Latin documents. Always acknowledge this and explain the threshold adjustment path.

DO NOT SAY: "It works at any scale"

SAY INSTEAD: Current implementation loads all chunks into memory. At 1M+ chunks this requires streaming mode, which is a FUTURE item. State the scaling limitation directly.

DO NOT SAY: "INSUFFICIENT_INPUT means the corpus is fine"

SAY INSTEAD: INSUFFICIENT_INPUT means the check could not run due to missing configuration. It is explicitly not PASS. A corpus with document_coverage INSUFFICIENT_INPUT has an unaudited coverage gap.

11.3 The One-Sentence Positioning Statement

DocIngestQA runs 11 deterministic checks on exported chunk JSONL files before indexing, catching structural defects -- missing pages, OCR noise, encoding corruption, near-duplicate chunks, poor split boundaries -- and producing PASS/WARN/FAIL status with chunk-level evidence, so RAG pipelines fail fast at ingestion rather than silently in production.

Appendix A -- Full Demo Walkthrough: Audit Corpus on 64-Chunk Corpus

A.1 Setup: Installing and Preparing the Demo

The demo corpus ships with the repo and contains 64 synthetic chunks across 10 source documents with deliberately seeded defects. To reproduce this walkthrough:

```
pip install docingestqa git clone https://github.com/SidharthKriplani/docingestqa cd docingestqa python examples/generate_demo_data.py # creates examples/assets/ python examples/audit_demo.py # runs the audit, writes outputs/
```

A.2 What generate_demo_data.py Produces

The generator creates two files: chunks.jsonl (64 chunk objects) and source_manifest.json (10 source documents with expected page counts). The injected defects are:

Defect	Injected Count	Expected Check to Fire
Missing source/page metadata	2 chunks	metadata_completeness WARN
Missing pages in manifest	10 pages across 2 docs	page_coverage FAIL
Short chunks (< 80 chars)	3 chunks	chunk_length WARN
OCR noise (U+FFFD, junk chars)	2 chunks	ocr_noise WARN
Exact duplicate chunk pairs	1 pair	duplicate_chunks WARN
Mojibake encoding sequences	2 chunks	encoding_health WARN
Poor split boundaries	4 chunks	split_quality WARN

A.3 Running the Audit

The audit_demo.py script instantiates IngestionAuditor with both the chunks and the manifest, runs all 11 checks, and writes the report in all three output formats:

```
from docingestqa import AuditConfig, IngestionAuditor auditor = IngestionAuditor(chunks_path='examples/assets/chunks.jsonl', documents_path='examples/assets/source_manifest.json', config=AuditConfig(), # all defaults ) report = auditor.run() report.to_json('outputs/ingestion_audit.json') report.to_markdown('outputs/ingestion_audit.md') report.to_html('outputs/ingestion_audit.html')
```

A.4 Actual Output (Verbatim from audit_demo.py)

Executive Summary:

Overall status : FAIL Chunks audited : 64 Sources : 10 Check counts : PASS=4 WARN=6 FAIL=1

Check-by-check results:

Check	Status	Summary
input_summary	PASS	64 chunks across 10 sources
metadata_completeness	WARN	2 chunks missing source or page metadata
document_coverage	PASS	0 missing documents, 0 orphan sources
page_coverage	FAIL	10 missing pages detected across 2 documents
chunk_length	WARN	3 short chunks (possible headers/fragments)
ocr_noise	WARN	2 chunks show OCR/extraction noise
duplicate_chunks	WARN	1 exact duplicate chunk pair
source_distribution	PASS	10 sources, largest at 14.1%
chunk_overlap	PASS	1 high-overlap consecutive pair (below threshold)
encoding_health	WARN	2 chunks contain mojibake sequences
split_quality	WARN	4 chunks with poor split boundaries

A.5 Interpreting the FAIL Finding: page_coverage

The FAIL on page_coverage is the most critical finding. The evidence in the JSON report identifies exactly which documents have missing pages:

```
{ "check": "page_coverage", "status": "FAIL", "issues": [ { "severity": "HIGH",  
  "message": "Document has pages missing from the chunk set.", "source":  
  "onboarding_guide.pdf", "evidence": {"missing_pages": [4, 5], "expected_pages": 8} }, {  
  "severity": "HIGH", "message": "Document has pages missing from the chunk set.",  
  "source": "technical_spec.pdf", "evidence": {"missing_pages": [3,4,5,6,7,8,9,10],  
  "expected_pages": 15} } ] }
```

Remediation: re-run the PDF extractor on onboarding_guide.pdf and technical_spec.pdf. Check whether pages 4-5 of onboarding_guide.pdf are scanned images (non-text pages) or text pages that failed to extract. Pages 3-10 of technical_spec.pdf suggest a systematic extraction failure -- possibly a password-protected section or a corrupt PDF range.

A.6 Interpreting the WARN Findings

The 6 WARN findings are below the FAIL threshold but represent real quality issues. The recommended review order (highest to lowest risk for retrieval quality):

1st: encoding_health WARN

Mojibake in 2 chunks destroys embeddings for those passages. Re-extract the source documents with explicit UTF-8 encoding.

2nd: ocr_noise WARN

2 chunks with U+FFFD replacement characters. Check whether the source pages are scanned images. Re-OCR with a higher-resolution scanner or enable image preprocessing in the extraction pipeline.

3rd: split_quality WARN

4 chunks with poor boundaries. Check chunker configuration: are sentence boundaries respected? Adjust splitter to use NLTK sentence tokenizer or spaCy sentence segmentation.

4th: duplicate_chunks WARN

1 exact duplicate pair -- minor. Deduplicate by chunk hash before indexing.

5th: chunk_length WARN

3 short chunks -- likely section headers or footers. Add a pre-processing step to filter out chunks below 80 chars.

6th: metadata_completeness WARN

2 chunks missing source/page. Minor -- likely from a source that doesn't export metadata. Add metadata injection in the extraction pipeline.

Appendix B -- AuditConfig Reference: Every Parameter Explained

Parameter	Type	Default	Effect if Too Low	Effect if Too High
min_chunk_chars	int	80	Short but legitimate chunks (headers) flagged WARN	All short chunks missed; fragment noise enters index
max_chunk_chars	int	1500	Long chunks not flagged; embedding truncation risk	Normal paragraphs flagged as too long
noisy_text_ratio_threshold	float	0.05	More OCR noise gets through	False positives on mathematical/code text
replacement_char_threshold	int	3	More U+FFFD noise passes	Single-char encoding errors trigger WARN
ngram_size	int	5	More false positive near-dupes (short n-grams match common phrases)	Near-dupes requiring long sequence match may be missed
near_duplicate_jaccard_threshold	float	0.85	More near-dupes caught (higher recall, lower precision)	Only very high overlap caught (lower recall)
fail_duplicate_pair_count	int	30	FAIL fires earlier on fewer duplicate pairs	FAIL fires later; many duplicates needed before block
overlap_jaccard_warn_threshold	float	0.40	More overlap pairs flagged WARN	Only high-overlap pairs flagged
overlap_jaccard_fail_threshold	float	0.70	FAIL fires on moderate overlap	Only extreme overlap triggers FAIL
mojibake_fail_rate	float	0.20	FAIL fires on lower mojibake rate (stricter)	Encoding issues can be widespread before FAIL triggers
fail_bad_split_rate	float	0.30	FAIL fires earlier on split quality	Many bad splits needed before FAIL triggers

Appendix C -- Output Schema Reference (v0.2)

Every output format (JSON, Markdown, HTML) contains the same underlying data. The JSON schema is the canonical representation:

```
{ "schema_version": "0.2", "metadata": { "docingestqa_version": "0.2.0", "generated_at": "2025-05-16T13:49:00Z", "chunk_count": 64 }, "executive_summary": { "overall_status": "FAIL", "chunk_count": 64, "check_counts": { "PASS": 4, "WARN": 6, "FAIL": 1 } }, "checks": [ { "check": "page_coverage", "status": "FAIL", "summary": "10 missing pages detected.", "metrics": { "missing_page_total": 10 }, "issues": [ { "check": "page_coverage", "severity": "HIGH", "message": "Document has pages missing.", "source": "onboarding_guide.pdf", "evidence": { "missing_pages": [4, 5], "expected_pages": 8 } } ], "recommendation": "Inspect parser logs for missing pages." } ], "interpretation_note": "DocIngestQA reports deterministic ingestion quality signals..." }
```

Key schema fields to reference in interviews:

Field	Where to Find It	Use Case
overall_status	executive_summary.overall_status	CI gate: exit code mapping
check_counts	executive_summary.check_counts	Dashboard summary metrics
issues[].severity	checks[].issues[].severity	Prioritize HIGH findings first
issues[].evidence	checks[].issues[].evidence	Chunk-level remediation context
interpretation_note	Root level	Must appear in all stakeholder reports
metrics	checks[].metrics	Quantitative context for each check finding